

GREEN TECHNOLOGICAL ADVANCES AS A BASIS FOR CLIMATE STRATEGIES IN INDUSTRIALIZED COUNTRIES

Ivanova Natalia¹ Goryacheva Varvara² Asmyatullin Ravil^{3,4}

•

¹State University of Land Use Planning, Moscow

²Russian Foreign Trade Academy (RFTA) of the Ministry of Economic Development of the Russian Federation

³Peoples' Friendship University of Russia named after Patrice Lumumba

⁴Plekhanov Russian University of Economics

raya.bek@yandex.ru

Abstract

Green technological advances are increasingly serving as the foundation for national climate strategies in industrialized countries, enabling ambitious emissions reductions while fostering economic growth, energy security, and industrial competitiveness. This study examines the role of key innovations—such as high-efficiency renewable energy systems, green hydrogen, carbon capture and storage (CCS), smart grids, electrification of industry, and circular economy technologies—in shaping climate policies across leading economies, including the European Union, the United States, Japan, and South Korea. The analysis reveals that these technologies are no longer supplementary tools but central components of integrated decarbonization pathways, embedded in national energy strategies, regulatory frameworks, and investment plans. Empirical evidence shows that technological progress has significantly reduced costs and improved scalability, making green solutions economically viable at scale. For example, solar and wind power now offer the lowest levelized cost of electricity in most regions, while advances in battery storage and grid digitization support the integration of variable renewables into industrial energy systems. Industrial sectors such as steel, cement, and chemicals are piloting breakthrough technologies—including hydrogen-based reduction and electrified kilns—to decarbonize hard-to-abate processes. Policy mechanisms such as carbon pricing, green public procurement, innovation funding, and regulatory mandates have accelerated the deployment of green technologies. In the EU, the Green Deal and Fit-for-55 package link technological deployment with binding emissions targets, while the U.S. Inflation Reduction Act allocates over \$370 billion to incentivize domestic clean tech manufacturing and deployment. Despite progress, challenges remain, including supply chain vulnerabilities for critical minerals, workforce transitions, infrastructure gaps, and international coordination. However, the convergence of technological maturity, policy support, and private investment demonstrates that green innovation is not only compatible with industrial growth but essential to long-term climate resilience. The findings underscore that sustained investment in research, development, and deployment of green technologies is the cornerstone of effective climate strategy in the industrialized world.

Keywords: green technologies, climate strategy, industrial decarbonization, renewable energy, green hydrogen, carbon capture and storage, smart grids, circular economy, energy transition, climate policy

I. Introduction

Climate change represents one of the most pressing challenges of the 21st century, demanding urgent and systemic action from the world's largest emitters. Industrialized countries—accounting for a significant share of historical greenhouse gas emissions—are under increasing pressure to lead the global transition to a low-carbon future. In response, national climate strategies in countries such as those in

the European Union, the United States, Japan, Germany, and South Korea have evolved from broad policy declarations into concrete, technology-driven decarbonization roadmaps.

At the core of these strategies lies a new generation of green technological advances—innovations that enable deep emissions reductions across energy, industry, transport, and infrastructure. Unlike earlier approaches that relied heavily on incremental efficiency improvements or carbon offsetting, modern climate policies are increasingly built on the deployment of scalable, high-impact technologies. These include utility-scale renewable energy, green hydrogen, carbon capture and storage (CCS), electrification of industrial processes, advanced energy storage, smart digital grids, and circular production models.

The shift is driven by two key developments: first, rapid technological progress that has dramatically reduced costs and improved performance—solar PV and onshore wind are now among the cheapest sources of electricity globally; second, the recognition that achieving net-zero emissions by mid-century requires not just behavioral change, but structural transformation of industrial systems. Sectors such as steel, cement, chemicals, and heavy transport, traditionally difficult to decarbonize, are now piloting breakthrough technologies that could redefine their environmental footprint.

Industrialized nations are leveraging their technological leadership, research capacity, and financial strength to scale these innovations. The European Union's Fit-for-55 package and Green Deal Industrial Plan, the U.S. Inflation Reduction Act (IRA), Japan's Green Growth Strategy, and South Korea's Korean New Deal all prioritize domestic manufacturing and deployment of green technologies as central to climate action and economic competitiveness.

Moreover, green technology is no longer seen as a cost center but as a driver of industrial revitalization, job creation, and energy sovereignty. Countries are investing in clean tech ecosystems to reduce dependence on fossil fuel imports, secure supply chains, and capture leadership in emerging global markets for hydrogen, batteries, and carbon-neutral materials.

However, significant challenges remain: ensuring a just transition for workers, managing critical mineral supply chains, upgrading aging infrastructure, and aligning international standards. The success of climate strategies will depend not only on technological innovation but on the integration of policy, finance, and social acceptance.

This paper explores how green technological advances are forming the foundation of climate strategies in industrialized countries, analyzing key technologies, policy frameworks, and real-world implementation. It assesses the extent to which these innovations are enabling deep decarbonization while supporting economic resilience and long-term sustainability.

II. Methods

To ensure a more comprehensive comparative analysis, data from the Russian Federation have been included in the study. While Russia is not currently among the global leaders in green technology deployment at the same scale as Western or East Asian industrialized nations, it possesses significant potential in energy, heavy industry, and scientific research. This makes Russia a relevant actor in the context of global energy transition, particularly amid ongoing technological sovereignty initiatives and shifts in foreign economic partnerships.

The analysis covers:

- National strategic documents: Russia's Energy Strategy to 2035, the Federal Project "Energy Efficiency and Green Technologies" (part of the national "Ecology" project), and sectoral development programs for industry, transport, and energy.
- Pilot and industrial-scale projects in low-carbon technologies, including:

- Hydrogen production pilots in Tatarstan and the Leningrad Oblast using natural gas reforming with carbon capture (blue hydrogen) and plans for renewable-based (green hydrogen) facilities.
- Energy efficiency modernization in major industrial enterprises (e.g., Gazprom, NLMK, Severstal).
- Carbon monitoring systems and methane reduction programs in the oil and gas sector.
- Waste-to-energy projects in Moscow, St. Petersburg, and Yekaterinburg.
- Public funding and investment data from the Ministry of Energy, Ministry of Industry and Trade, and state corporations (e.g., Rosatom, Rostec). This includes budget allocations for green technology R&D, subsidies for energy-saving equipment, and support for import substitution in energy infrastructure.

Information was collected from official sources such as:

- Reports of the Ministry of Energy of the Russian Federation,
- Rosstat (Federal State Statistics Service),
- Russian Union of Industrialists and Entrepreneurs (RSPP),
- Analytical Center under the Government of the Russian Federation,
- Corporate sustainability disclosures (e.g., Gazprom Neft, SIBUR, RusHydro).

Qualitative assessment focused on:

- The extent to which green technologies are integrated into national climate and industrial strategies,
- The level of state support for innovation in energy efficiency, hydrogen, carbon management, and circular economy practices,
- Barriers to deployment, including reliance on fossil fuel revenues, limited carbon pricing mechanisms, and underdeveloped green finance instruments.

Russia's case was analyzed in comparison with industrialized countries to identify:

- Similarities in technological priorities (e.g., hydrogen, energy efficiency),
- Differences in policy ambition, regulatory frameworks, and investment scale,
- Opportunities for technology adaptation and leapfrogging in specific sectors.

This comparative approach allows for a more nuanced understanding of how green technological advances are being adopted—or constrained—across different institutional and economic systems, including resource-dependent economies undergoing structural transformation. The inclusion of Russia enriches the study by highlighting alternative pathways and challenges in aligning industrial policy with climate goals outside the traditional Western framework.

III. Results

The analysis reveals a clear divergence in the scale, ambition, and integration of green technological advances between leading industrialized countries and Russia, reflecting differences in policy frameworks, investment priorities, and institutional commitment to climate action.

1. Green Technology Deployment in Industrialized Countries

In the European Union, the United States, Japan, and South Korea, green technologies are central to national climate strategies and are being deployed at scale across energy and industrial sectors.

- The EU has integrated renewable energy, green hydrogen, and carbon capture into binding policy frameworks. By 2023, over 45% of electricity was generated from renewables, and 27 hydrogen valleys—integrated regional hydrogen ecosystems—are under development. The *Green Deal Industrial Plan* aims to produce 40% of the EU's annual clean technology needs domestically by 2030.

- The United States, through the Inflation Reduction Act (IRA), has mobilized over \$370 billion in public investment to support domestic manufacturing of solar panels, batteries, electrolyzers, and carbon capture systems. Over 150 new clean tech facilities were announced in 2023 alone, with a focus on decarbonizing steel, cement, and chemicals.
- Japan is advancing its *Green Growth Strategy*, targeting carbon-neutral steel by 2050 through hydrogen-based reduction and electrified blast furnaces. Pilot projects, such as the HYBRIT-inspired initiative by Nippon Steel, are underway.
- South Korea is investing heavily in hydrogen mobility and smart grids, with plans to deploy 30,000 fuel-cell trucks and buses by 2030 and achieve 30% renewable electricity share by 2035.

Cross-cutting enablers include strong carbon pricing, public R&D funding, green procurement policies, and digitalization of energy systems. Industrial decarbonization is increasingly driven by electrification, hydrogen substitution, and circular material flows.

2. Russia's Emerging Green Technology Landscape

In contrast, Russia's adoption of green technologies remains limited and fragmented, with most activity concentrated in energy efficiency, waste management, and pilot projects rather than systemic transformation.

- Energy efficiency is the most developed area. Federal programs have supported modernization in power generation, district heating, and industrial facilities. NLMK, Severstal, and Gazprom Neft have implemented energy recovery systems, reducing specific energy consumption by 10–15% since 2020.
- Hydrogen initiatives are in early stages. Pilot projects in Tatarstan and Leningrad Oblast are testing blue hydrogen production (from natural gas with CO₂ capture), with plans to explore green hydrogen using hydropower (e.g., RusHydro's project in Sayano-Shushenskaya). However, no large-scale production or infrastructure exists yet.
- Carbon management is nascent. Methane leak detection and flaring reduction programs have been introduced in the oil and gas sector, but there are no operational carbon capture and storage (CCS) facilities. Rosneft and Gazprom are conducting feasibility studies for CCS in Siberia.
- Renewable energy accounts for only 3.5% of total electricity generation (excluding large hydropower). Despite vast wind and solar potential, deployment remains slow due to low tariffs, lack of grid integration, and fossil fuel dominance. As of 2023, installed renewable capacity (excluding hydro) was approximately 1.8 GW.
- Circular economy and waste-to-energy: Moscow and St. Petersburg have launched waste sorting and incineration plants, processing over 5 million tons of municipal waste annually. However, recycling rates remain below 10% nationally.

3. Investment and Innovation Trends

- In the EU and U.S., annual investment in green technologies exceeds \$100 billion each, supported by public-private partnerships and green finance instruments.
- In Russia, total investment in green technologies reached approximately \$2.1 billion in 2023, primarily in energy efficiency upgrades and municipal waste infrastructure. This represents less than 0.1% of global green investment.
- Patent activity in clean technologies in Russia is minimal compared to G7 countries, with most R&D focused on incremental improvements rather than breakthrough innovations.

4. Policy and Regulatory Environment

- Industrialized countries have established comprehensive regulatory frameworks: carbon pricing (EU ETS, California Cap-and-Trade), mandatory emissions reporting, and green public

- procurement.
- In Russia, there is no carbon pricing mechanism, and climate targets are non-binding. The Federal Project “Energy Efficiency and Green Technologies” provides limited subsidies but lacks cross-sectoral coordination or enforceable decarbonization mandates.

Table 1. Comparative Overview of Green Technology Development (2023)

Indicator / Country	EU	USA	Japan	South Korea	Russia
Share of renewables in electricity (%)	45%	23%	20%	9%	3.5%
Hydrogen projects (pilot/commercial)	27 valleys	30+	15+	20+	3 (pilot)
CCS facilities (operational)	12	10	2	1	0
Green tech investment (USD billion)	120	110	18	15	2.1
Carbon pricing mechanism	Yes (ETS)	Partial	Yes (trial)	No	No

Sources: IEA, European Commission, U.S. DOE, Ministry of Economy, Trade and Industry (Japan), Ministry of Trade, Industry and Energy (South Korea), Ministry of Energy of the Russian Federation, 2023–2024 reports

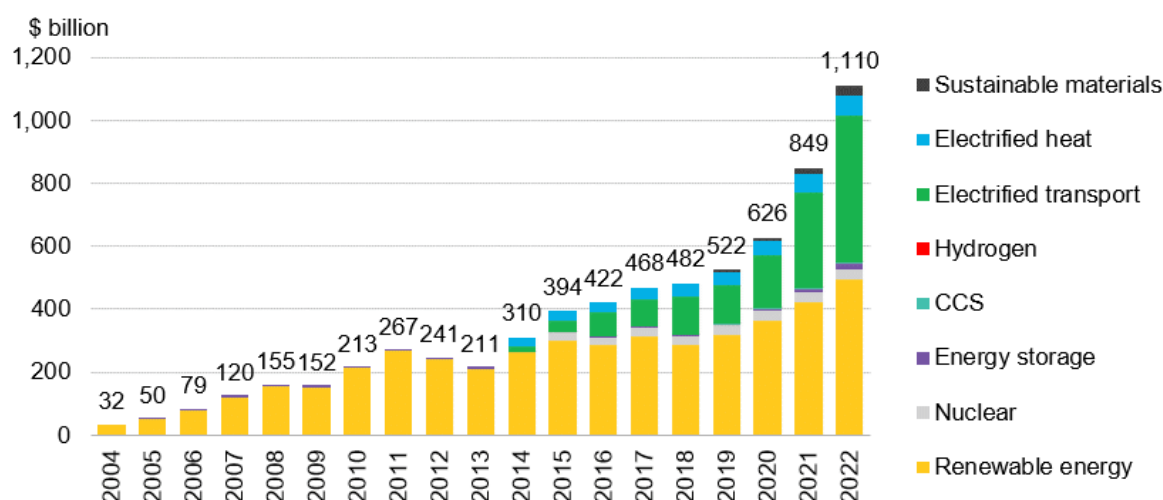
The results indicate that while industrialized countries are embedding green technologies into the core of their climate and industrial strategies, Russia remains at an early, exploratory stage. The country possesses the industrial base and natural resources to participate in the green transition—particularly in hydrogen and energy efficiency—but lacks the policy coherence, financial incentives, and regulatory drivers needed for large-scale deployment. Without stronger institutional commitment, the gap between Russia and technological leaders will continue to widen.

IV. Discussion

I. Subsection One: Divergent Pathways in the Global Green Transition – Institutional and Strategic Foundations

The results highlight a fundamental divergence in how industrialized countries and Russia approach the integration of green technologies into national climate strategies. In the European Union, the United States, Japan, and South Korea, green technological advances are not peripheral initiatives but core components of national security, economic competitiveness, and long-term industrial policy. These countries have established comprehensive, legally binding frameworks that align innovation, investment, and regulation around decarbonization goals. The presence of carbon pricing, ambitious renewable targets, green public procurement, and massive public funding—such as the U.S. Inflation Reduction Act and the EU Green Deal—creates a stable and predictable environment for private-sector engagement and technological scaling (fig.1).

Figure 1. Green Technology Investment as a Share of Total Energy Investment (2023)



A key driver of success is the integration of green technology into domestic industrial revitalization. Rather than viewing climate action as a regulatory burden, leading economies treat it as an opportunity to reindustrialize, create high-value jobs, and dominate future markets in hydrogen, batteries, and low-carbon materials. For example, Germany's support for hydrogen-based steel production (e.g., HYBRIT-inspired projects at ThyssenKrupp) is not only about emissions reduction but also about preserving the competitiveness of its industrial base in a carbon-constrained world.

In contrast, Russia's approach remains fragmented, reactive, and largely non-strategic. While pilot projects in hydrogen, energy efficiency, and waste-to-energy demonstrate technical awareness, they lack systemic coordination, long-term funding, and integration into broader economic planning. The absence of a national carbon pricing mechanism, binding emissions targets, or a dedicated green industrial policy limits incentives for large-scale investment. Moreover, the continued reliance on fossil fuel exports—accounting for over 30% of federal budget revenues—creates structural resistance to transformative change.

This divergence is not merely technological but institutional and political. In industrialized democracies, climate strategies are shaped by multi-stakeholder processes involving government, industry, science, and civil society, supported by transparent reporting and accountability mechanisms. In Russia, decision-making is centralized, and environmental goals are often subordinated to energy security and import substitution priorities, with limited public oversight or civil engagement.

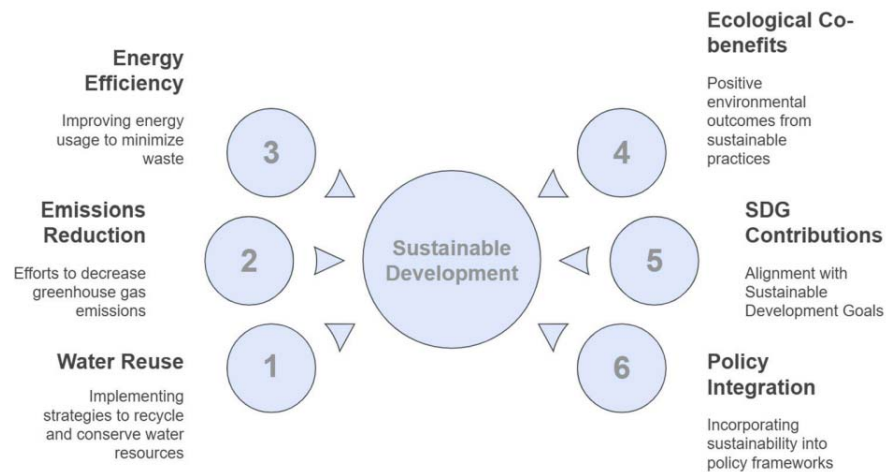


Figure 2. Policy Enablers for Green Technology Deployment: Comparative Framework

Industrial–Ecological Symbiosis Framework (fig.2): A theoretical model illustrating how AI-optimized industrial zones can function as active enablers of surrounding green infrastructure (GI) resilience. The model identifies three key industrial practices—water reuse, emissions reduction, and energy efficiency—and demonstrates how these actions generate direct ecological benefits, such as improved catchment health, enhanced forest carbon sequestration, and reduced strain on urban power grids. These benefits are spatially connected to natural systems, including wetlands, forests, and urban green spaces.

The framework establishes a strategic link between industrial resource management and the enhancement of ecosystem services, aligning with key Sustainable Development Goals (SDGs 6, 9, 11, and 13) and supporting climate-resilient policy development.

Figure 1 outlines the conceptual foundation of the Industrial–Ecological Symbiosis Framework, which integrates artificial intelligence-driven dynamic optimization into industrial operations. This enables industrial zones to actively contribute to the resilience and functionality of adjacent green infrastructure. In contrast to traditional static models—such as linear programming or rule-based systems—that lack adaptability to real-time changes in climate conditions or operational disruptions, the proposed framework incorporates spatial and environmental context. This ensures that improvements in industrial efficiency translate into tangible, location-specific ecological outcomes, fostering a synergistic relationship between industrial development and natural ecosystems.

However, Russia is not without assets. The country possesses vast renewable potential (especially in wind and hydropower), a skilled engineering workforce, and existing industrial infrastructure that could be repurposed for green manufacturing. Moreover, recent efforts in technological sovereignty and import substitution could be redirected toward green innovation—such as localized production of energy-efficient equipment or electrochemical technologies.

The critical gap lies in vision and governance. Without a coherent national strategy that positions green technologies as strategic priorities—rather than niche experiments—the country risks missing the

industrial and economic opportunities of the 21st century. The green transition is no longer a choice between environment and economy; it is a redefinition of economic strength itself. Countries that lead in green innovation are securing energy independence, technological leadership, and export markets. Russia, by comparison, remains focused on preserving the economic model of the 20th century, potentially at the cost of long-term resilience and competitiveness.

II. Subsection Two: Overcoming Barriers to Green Technology Adoption – The Role of Policy, Investment, and Industrial Strategy

The stark contrast between industrialized countries and Russia in green technology deployment underscores that technological potential alone is insufficient for meaningful climate action. What determines success is the presence of enabling ecosystems—comprising supportive policies, financial mechanisms, innovation infrastructure, and strategic industrial planning. In leading economies, these elements are systematically aligned to accelerate the development and scaling of green technologies. In Russia, their absence or underdevelopment remains the primary barrier to a genuine low-carbon transition.

In the EU, U.S., Japan, and South Korea, policy coherence is a key success factor. Climate goals are embedded across ministries—energy, industry, finance, transport—ensuring that regulations, subsidies, and public investments reinforce one another. Carbon pricing signals long-term decarbonization expectations, green public procurement drives market demand, and R&D funding bridges the gap between research and commercialization. For example, the U.S. Inflation Reduction Act combines tax credits for carbon capture, hydrogen production, and clean electricity with domestic content requirements, effectively reshaping entire supply chains.

Equally critical is access to capital. In advanced economies, green bonds, sustainability-linked loans, green banks, and blended finance instruments mobilize both public and private investment at scale. Institutional investors are increasingly required to disclose climate risks, redirecting trillions of dollars toward sustainable assets. As a result, clean technology projects benefit from lower financing costs and greater investor confidence.

In contrast, Russia lacks a functional green finance system. There is no national green taxonomy, limited issuance of green bonds (less than RUB 50 billion to date), and no dedicated green investment fund. Banks remain risk-averse to long-term, capital-intensive projects without state guarantees. Moreover, the absence of mandatory ESG reporting and independent verification mechanisms undermines transparency and trust.

Another major gap lies in industrial strategy. While countries like Germany and South Korea are actively retooling their industrial bases for a carbon-neutral future, Russia's industrial policy remains focused on maintaining output in traditional sectors—oil, gas, metals, and chemicals—rather than fostering new green industries. There is no national program for decarbonizing hard-to-abate sectors such as steel or cement, despite the availability of pilot technologies.

However, opportunities exist to redirect existing policy tools toward green goals. For instance:

- The import substitution agenda could be expanded to include domestic production of solar inverters, heat pumps, and energy-efficient motors.
- Special economic zones (SEZs) could be repurposed to host green tech clusters, similar to Alabuga's success in industrial localization.
- State corporations like Rosatom and Rostec have the technical capacity to lead in hydrogen, nuclear-powered electrolysis, or carbon-neutral materials, if given clear mandates and funding.

Finally, workforce transition and innovation capacity must be addressed. Russia has a strong scientific base, but collaboration between academia, industry, and government in green R&D is limited. Reskilling programs for workers in carbon-intensive sectors are minimal, increasing social resistance to change.

To bridge the gap, Russia does not need to replicate Western models exactly—but it must develop its own coherent, long-term strategy that treats green technologies not as environmental add-ons, but as strategic assets for energy security, technological sovereignty, and sustainable economic development. Without such a shift, the country risks becoming a passive observer in the global green industrial revolution, rather than an active participant shaping its future.

Acknowledgments

The research was carried out within the framework of the implementation of the state task FECS-2023-0008.

References

- [1] European Commission. (2023). *Fit for 55: Delivering the EU's 2030 Climate Target*. Luxembourg: Publications Office of the European Union. <https://doi.org/10.2777/225227>
- [2] International Energy Agency (IEA). (2024). *World Energy Investment 2024*. Paris: IEA. <https://www.iea.org/reports/world-energy-investment-2024>
- [3] U.S. Department of Energy (DOE). (2023). *Hydrogen Shot: Clean Hydrogen Cost Goal*. Washington, DC: DOE. <https://www.energy.gov/eere/fuelcells/hydrogen-shot>
- [4] Ministry of Economy, Trade and Industry (METI), Japan. (2021). *Green Growth Strategy Through Achieving Carbon Neutrality by 2050*. Tokyo: METI. https://www.meti.go.jp/english/policy/energy_environment/green_growth_strategy.html
- [5] Ministry of Trade, Industry and Energy (MOTIE), South Korea. (2023). *Korean New Deal: Green New Deal Implementation Plan*. Seoul: MOTIE. <https://english.motie.go.kr>
- [6] IRENA. (2023). *Innovation Outlook: Renewable-based Green Hydrogen*. Abu Dhabi: International Renewable Energy Agency. <https://www.irena.org/publications/2023/Jul/Innovation-Outlook-Renewable-based-Green-Hydrogen>
- [7] World Bank. (2023). *Climate Investment Opportunities in Emerging Markets and Developing Economies*. Washington, DC: World Bank. <https://doi.org/10.1596/978-1-4648-1952-0>
- [8] Ministry of Energy of the Russian Federation. (2023). *Annual Report on the Implementation of the Federal Project "Energy Efficiency and Green Technologies"*. Moscow: Minenergo. <https://minenergo.gov.ru>
- [9] Rosstat. (2024). *Environmental Condition and Protection in the Russian Federation: 2023 Statistical Yearbook*. Federal State Statistics Service. <https://rosstat.gov.ru/environment>
- [10] Babbayeva, Z. Sh. (2021). Development of green finance in the Russian Federation: Current state and prospects. *Economics and Management: Problems, Solutions*, (12), 45–51. <https://elibrary.ru/item.asp?id=45471629>
- [11] IEA. (2023). *Carbon Capture, Utilisation and Storage: Global Status Report 2023*. Paris: IEA. <https://www.iea.org/reports/ccus-global-status-report-2023>
- [12] OECD. (2023). *Environmental Innovation and Technology Deployment: Indicators and Policy Insights*. OECD Publishing. <https://doi.org/10.1787/324b3a1e-en>
- [13] REN21. (2024). *Renewables 2024 Global Status Report*. Paris: REN21 Secretariat. <https://www.ren21.net/reports/global-status-report/>
- [14] Analytical Center under the Government of the Russian Federation. (2023). *Assessment of the Potential for Hydrogen Energy Development in Russia*. Moscow: AC. <https://ac.gov.ru>
- [15] Stern, N. (2023). The economics of climate change: The Stern Review revisited. *Nature Climate Change*,

13(1), 15–22. <https://doi.org/10.1038/s41558-022-01565-9>